UG CAMPUS SERVICE OPTIMIZER

PROJECT PROGRESS REPORT

Smart Maintenance Request Management and Algorithmic Route Optimization for the University of Ghana

Course: DCIT 204 / DCIT 308 Data Structures and Algorithms
Institution: Department of Computer Science, University of Ghana, Legon
Group Leader: Kelvin Agyare Yeboah | ID: 22159683 | GitHub: kelvinagyareyeboah
Date: August 18, 2026
Status: Core Implementation Phase Complete | 100% DSA Modules and UI Visualizer Operational

1. Executive Summary

The UG Campus Service Optimizer is an intelligent algorithmic facility management system designed specifically for the University of Ghana, Legon campus. The system addresses important operational challenges in university maintenance, including emergency faults such as power outages and pipe bursts, efficient routing of repair vehicles across campus landmarks, and maximization of technician productivity under strict time constraints.

The team has successfully implemented the complete core DSA engine in Java 22 SE from first principles, without relying on built in Java data structure libraries. A React 18 and TypeScript interactive web interface has also been developed for real time visualization, demonstration, and empirical performance profiling.

2. System Architecture

Visual Frontend

React 18
TypeScript
Tailwind CSS
Leaflet

↓

Backend and Storage

Supabase
PostgreSQL Database

↓

Core DSA Engine

Java 22 SE
Decoupled Modular Architecture

2.1 Core Java DSA Engine

• Built decoupled Java modules implementing fundamental data structures and algorithms from first principles.

• Implemented memory management, pointer manipulation, and recursive balance operations without external utility libraries.

2.2 Interactive Visual Frontend

• Built distinct dashboard views corresponding to each core data structure and algorithm for real time visual verification during oral defense.

• Integrated Leaflet and OpenStreetMap with more than 50 University of Ghana campus landmarks and more than 100 road connections.

• Mapped locations including Balme Library, JQB, Sarbah Hall, Commonwealth Hall, UG Hospital, and the Department of Computer Science.

2.3 Empirical Performance Profiling Engine

• Developed an automated benchmarking suite measuring CPU execution time in nanoseconds and milliseconds.

• Records operation counts and throughput scaling from N = 100 to N = 10,000.

3. Team Member Roster and Assignment Table

No.	Student ID	Full Name	GitHub Username	Primary DSA Assignment	Tracking	Status
1	22159683	Kelvin Agyare Yeboah	kelvinagyareyeboah	Group Lead, System Architecture and Profiler Engine, PerformanceBenchmark.java, BTree.java	Core Integration	Completed
2	22014087	Anthony Gudu	agudu50	CustomQueue.java, FIFO Queue	Core Module	Completed
3	22152286	Boadu Acheampong Yaw Asante	yawasante dev	GreedyTaskScheduler.java, Greedy Choice	Core Module	Completed
4	22046873	Adjei David Boafo	DavidAdjeiB	CustomPriorityQueue.java and DisjointSet.java, Union Find	Issue 23 / PR 28	Completed
5	22099559	Osman Ilyas	samuncleML	DynamicProgrammingOptimizer.java, 0/1 Knapsack DP	Core Module	Completed
6	22027295	Frank Tsetse Kofi Bless	Tsetse bless	CustomStack.java, Audit Trail Stack	Core Module	Completed
7	22306887	AFFADZI, GADDIEL DONALD	a pssh	BFSExplorer.java, Breadth First Search	Core Module	Completed
8	22372832	MAGLO, PRIMUS NORVINYO	Phoenicx1	DFSConnectivity.java and CustomHashTable.java	Core Module	Completed
9	22335736	DEH, ARNOLD KWASI	arnoldmackenzie	CampusGraph.java, Adjacency Matrix and List	Issue 24	Completed
10	22301802	LUKMAN, HAMNA TIMTONI	Hamna T	DijkstraShortestPath.java, Dijkstra Algorithm	Core Module	Completed
11	22328690	DARKO, ELIZABETH ACHEAMPOMAA	CODEwithBETH dev	BinarySearchTree.java, BST Search Index	Core Module	Completed
12	22402808	AFRIYIE, GLORY OSEI WUSU	ry glo	PrimMST.java, Prim Minimum Spanning Tree	Core Module	Completed
13	22403283	ATTIPO, SHIKA ETHLYN	Ethlyn Codes	AVLTree.java, Self Balancing Tree	Core Module	Completed
14	22335990	SIMONS, SAMUEL WEABO JUNIOR	Simons990	KruskalMST.java and BruteForceScheduler.java	Issue 27 / PR 34	Completed

15	22413744	PARWANI, RAKESH RODNEY	rakeshparwani	BruteForceScheduler.java and Sorting Suite	Issue 27 / PR 34	Completed
16	22395054	AGYEMANG, BENEDICT NANA OBENG	NANAOBENG1234	BruteForceScheduler.java and Deque/CircularQueue	Issue 27 / PR 34	Completed

4. Implemented Data Structures Summary

Data Structure	Source Module	Campus Application	Complexity
FIFO Queue	CustomQueue.java	Submission order processing for routine maintenance	O(1) / O(1) / O(1)
Max Binary Heap	CustomPriorityQueue.java	Immediate preemption for high priority and emergency faults	O(1) / O(log n) / O(log n)
LIFO Audit Stack	CustomStack.java	Action log tracking, supervisor audit history, and undo operations	O(1) / O(1) / O(1)
Dynamic Resizable Array	CustomArrayList.java	Resizable list of service domains and campus landmarks	O(1) / O(1) amortized / O(n)
Doubly Linked List	CustomLinkedList.java	Dynamic collections of active maintenance teams and issues	O(1) / O(n) / O(n)
Separate Chaining Hash Table	CustomHashTable.java	Fast lookup for request IDs, emails, and locations	O(1) / O(1) / O(n)
Binary Search Tree	BinarySearchTree.java	Key sorted indexing of requests by timestamp and priority	O(log n) / O(log n) / O(n)
AVL Tree	AVLTree.java	Self balancing tree guaranteeing logarithmic performance	O(log n) / O(log n) / O(log n)
Multi way B Tree	BTree.java	Database style indexing for large historical request logs	O(log n) / O(log n) / O(log n)
Adjacency List and Matrix	CampusGraph.java	UG Legon road network representation	Space O(V + E) and O(V^2)

Disjoint Set	DisjointSet.java	Connectivity tracking with path compression and union by rank	O(1) / alpha(n) / alpha(n)
Fixed Circular Queue	CustomCircularQueue.java	Fixed memory telemetry buffer for vehicle tracking	O(1) / O(1) / O(1)
Double Ended Queue	CustomDeque.java	Dispatcher queue allowing supervisor front insertion	O(1) / O(1) / O(1)

5. Implemented Algorithms Summary

Algorithm	Source Module	Practical Problem Solved	Complexity

Dijkstra Algorithm	DijkstraShortestPath.java	Calculates shortest navigation distance and travel time for repair vans	O((V + E) log V)
0/1 Knapsack DP	DynamicProgrammingOptimizer.java	Maximizes shift priority points within technician working hour limits	O(N × W)
Greedy Choice Scheduler	GreedyTaskScheduler.java	Rapid dispatch using priority to time ratio	O(N log N)
Brute Force Scheduler	BruteForceScheduler.java	Evaluates all task combinations to demonstrate DP efficiency	O(2^N)
Prim MST	PrimMST.java	Minimum cost campus utility network planning	O(E log V)
Kruskal MST	KruskalMST.java	Minimum spanning network calculation using Disjoint Set	O(E log E)
Breadth First Search	BFSExplorer.java	Discovers landmarks within a specified number of road hops	O(V + E)
Depth First Search	DFSConnectivity.java	Validates reachability and detects network cycles	O(V + E)
Linear Search	LinearSearch.java	Unsorted ticket array scan with exact comparison counting	O(n)
Binary Search	BinarySearch.java	Fast search on sorted request arrays	O(log n)
Sorting Suite	SelectionSort, InsertionSort, MergeSort, QuickSort	Compares quadratic and logarithmic sorting performance	$O(n^2)$ versus O(n log n)

6. Empirical Performance Benchmark Results

Sample metrics generated by the built in Java performance engine, PerformanceBenchmark.java.

Campus Map Data

Graph Initialization: More than 50 locations and 101 road connections

Urgent Priority Preemption

Processing Order Based on Priority Weight

REQ 102: Emergency Priority, Power Fault at Main Server Room, Weight 4

REQ 105: Emergency Priority, Pipe Burst at Sarbah Hall, Weight 4

REQ 103: High Priority, Water Leakage at JQB, Weight 3

REQ 101: Medium Priority, Sanitation at Balme Library, Weight 2

Route Optimization

Route Search: Main Gate to UG Hospital

Optimal Path: Main Gate → JQB → Balme Library Junction → UG Hospital

Total Distance: 1,240 m

Estimated Travel Time: 4.2 minutes

Dynamic Programming Shift Optimization

Shift Capacity: 5.0 hours

Optimal Task Selection: 3 tasks selected

Total Priority Score: 11 points

Hours Used: 5 out of 5 hours

Performance Metrics

Metric	Result
FIFO Queue Operations	0.012 ms
Priority Queue Operations	0.028 ms
Hash Table Lookup	420 nanoseconds
Binary Search at N = 10,000	490 nanoseconds
Dijkstra Shortest Path	0.147 ms
0/1 Knapsack DP at N = 10	0.085 ms

Brute Force Comparison	12.4 ms

7. Recommended Points for TA Feedback

7.1 Map Edge Modifiers

Would you recommend adding dynamic weight adjustments for traffic during peak lecture hours on campus?

7.2 Benchmark Presentation Format

Are the nanosecond comparison tables sufficient, or should visual bar charts be included in the presentation slides?

7.3 Oral Defense Demonstration

Should team members defend their modules directly from the Java source code files or through the visual React dashboard?

8. Project Completion Statement

The core implementation phase of the UG Campus Service Optimizer is complete. The project currently combines a fully implemented Java DSA engine with an interactive React visualization layer and an empirical benchmarking system.

The remaining focus is on presentation refinement, validation, demonstration preparation, and incorporating feedback from the teaching assistant.

Prepared by: Kelvin Agyare Yeboah and Team
Course: DCIT 204 / DCIT 308 Data Structures and Algorithms
Department of Computer Science, University of Ghana, Legon